Inferential Statistics 23 May

Saturday, 23 May 2026 20:03

Data Science

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Data Analysis

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Machine Learning

- → SQL
- → Excel
- → Tableau / PowerBI
- → Statistics
- → Python / R

There are 2 branches of statistics :

① Descriptive Statistics :

② Inferential Statistics →

- → Confidence Interval
- → Hypothesis Testing

Higher the spread → More the uncertainity / variation

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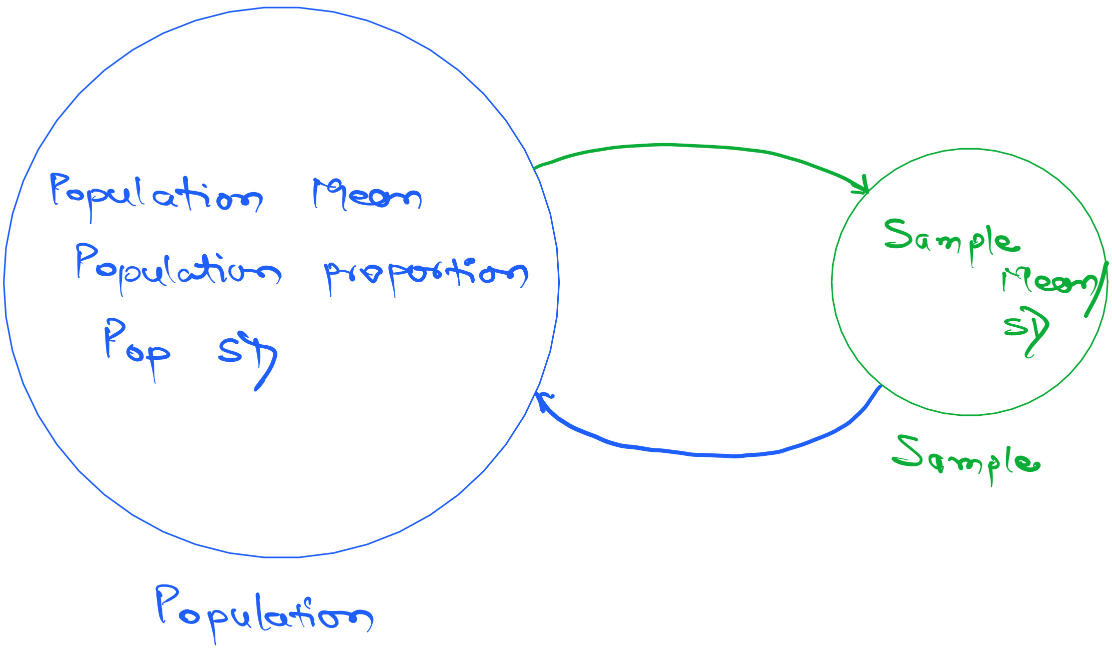
More Risk

Inferential Statistics

Estimating something which is unknown.



Budget Constraint

Time Constraint

Central Limit Theorem (CLT)

If you take 1000 samples from the pop, and calculate the sample mean of each sample, the sample means will be always normally distributed.

$S_1(5000)$ $S_2(5000)$ $S_3(5000)$ ---- $S_{1000}(5000)$

$\downarrow$ $\downarrow$ $\downarrow$ ---- $\downarrow$

$\bar{x}$ $\bar{x}$ $\bar{x}$ $\bar{x}$

$x_1 \quad x_2 \quad x_3 \quad x_{1000}$

$[\bar{x}_1, \bar{x}_2, \bar{x}_3 ---- \bar{x}_{1000}]$ 1000 SMs

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kdeplot

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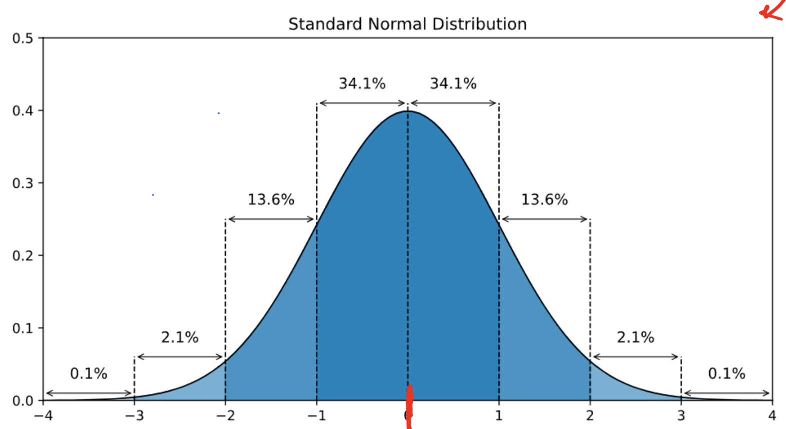


Mean of sample Means = Population Mean

PM — SM

Confidence Interval :

CI = Sample Mean ± Margin of Error

= 1.8 hrs/day ± 0.09 hrs

CI = 1.71 hrs/day to 1.89 hrs/day

PM

Why Confidence Interval :

Confidence Level	Interval
90%	1.8 hrs ± 0.04 hrs
95%	1.8 hrs ± 0.09 hrs
99%	1.8 hrs ± 0.12 hrs

Formula to calculate CI :

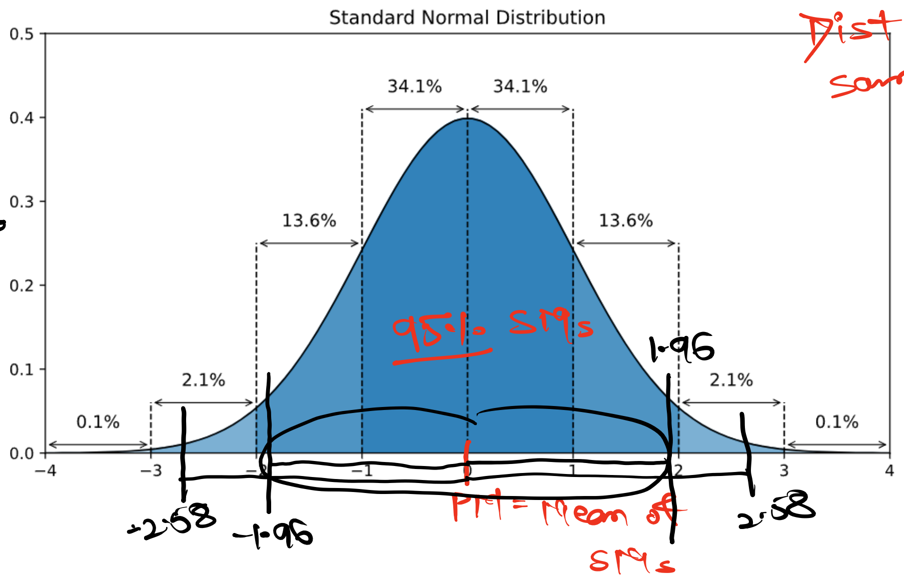
1000 SMs

CLT

95% SMs

950

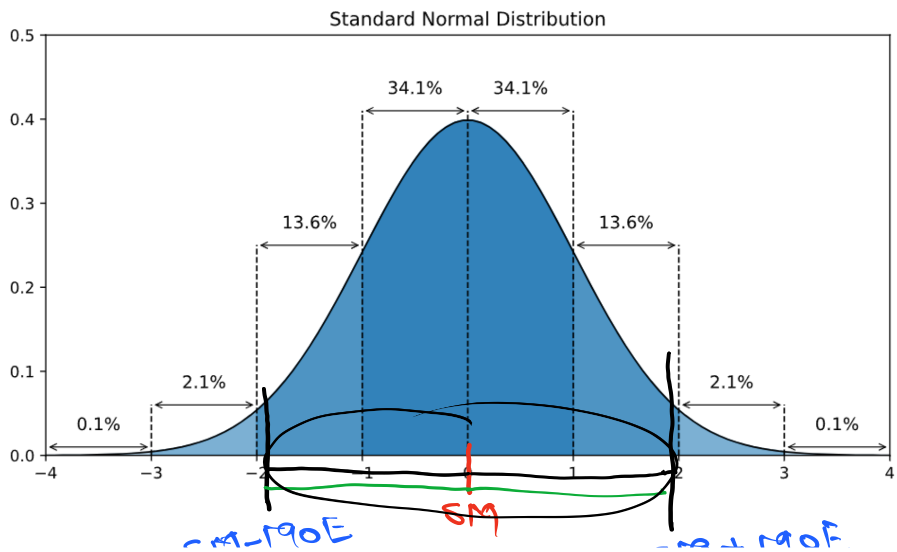
99% CI



Dist of sample means

$z^* = 1.96$

① Aim: Confidence Interval



$\bar{x} - 1.96 \times SE$ $\bar{x} + 1.96 \times SE$

① 1-sample

② SM

$$\bar{x} \pm z^* \times \frac{S}{\sqrt{n}}$$

$\bar{x} - 1.96 \times SE$ to $\bar{x} + 1.96 \times SE$

Seagate / WD ⟶ HDDs / SSDs

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Decide optimal warranty duration

Sample of 200 HDDs

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SM = 5.2 yrs

5 yrs

SM ± MOE

5.2 yrs ± 0.8 yrs

4.4 yrs to 6 yrs

$\boxed{\text{4 yrs}}$